Ideation Phase

Empathize & Discover

Date	08 March 2026
Team ID	SmartBridge-AI-01
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Empathy Map Canvas

User Persona: Insurance Investigator / Claims Analyst

THINK & FEEL

Worried about missing fraudulent patterns | Frustrated with manual review processes | Hopes for a reliable automated system | Concerned about false positives blocking genuine claims

HEAR

- Colleagues reporting high fraud rates in vehicle insurance
- Management pushing for faster claim approvals
- Industry experts talking about AI-driven fraud detection
- News about rising insurance fraud costing billions annually

SEE

- Hundreds of insurance claims arriving daily
- Excel spreadsheets and manual checklists being used
- Colleagues spending hours reviewing each claim
- Fraudulent patterns that are hard to spot manually

SAY & DO

- *'We need a smarter way to flag suspicious claims before they're approved'*
- Manually cross-references claim data with historical fraud patterns
- Uses intuition and experience to flag unusual cases
- Requests additional documentation from claimants that seem suspicious

PAIN

- Fear of approving fraudulent claims undetected
- Frustration with time-consuming manual investigation
- Inconsistency in fraud detection across team members
- Lack of data-driven tools to support decisions

GAIN

- Accurate, real-time fraud prediction with ML models
- Reduced processing time per claim
- Consistent and auditable fraud detection process
- Confidence in data-backed decisions

Ideation Phase

Brainstorm & Idea Prioritization

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Step 1 – Problem Statement

How might we help insurance investigators accurately and efficiently detect fraudulent claims using AI, so that genuine claimants are protected and company losses are minimized?

Step 2 – Brainstormed Ideas

Idea	Category
Train ML classifier (Random Forest) to predict fraud probability	Machine Learning
Use Decision Tree for interpretable fraud rules	Machine Learning
Apply KNN to detect anomalous claim patterns	Machine Learning
Gradient Boosting for high-accuracy ensemble model	Machine Learning
EDA on claim features: age, premium, incident severity	Data Analysis
Feature engineering: derive fraud-risk indicators	Data Analysis
Real-time claim scoring API for integration with CRM	System Integration
Dashboard for investigators showing flagged claims	User Interface
Explainability module (SHAP values) for each prediction	Transparency
Auto-alert system when fraud probability exceeds threshold	Automation
Feedback loop: investigator verdict feeds back to retrain model	Model Improvement
Secure role-based access for investigators and admins	Security

Step 3 – Idea Prioritization (Impact vs Feasibility)

Idea	Impact (H/M/L)	Feasibility (H/M/L)	Priority

Random Forest Fraud Classifier	High	High	1st
EDA & Feature Engineering	High	High	2nd
Real-time Scoring API	High	Medium	3rd
Investigator Dashboard	Medium	Medium	4th
SHAP Explainability	Medium	Medium	5th
Decision Tree (Interpretable Rules)	High	High	2nd
Feedback Retraining Loop	Medium	Low	6th
Auto-Alert System	High	Medium	3rd

Ideation Phase

Define the Problem Statements

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Customer Problem Statement

PS No.	I am (Customer)	I'm trying to...	But...	Because...	Which makes me feel
PS-1	An insurance claims investigator	Quickly and accurately identify fraudulent insurance claims	I must manually review hundreds of claims with no consistent tool	There is no AI-driven system integrated into our workflow	Overwhelmed, anxious, and unsure of my decisions
PS-2	An insurance company manager	Reduce financial losses caused by approved fraudulent claims	Detection rates remain low and inconsistent across the team	Manual review lacks data-driven pattern recognition capabilities	Concerned, responsible for large undetected fraud losses

Project Design Phase II

Solution Requirements (Functional & Non-Functional)

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Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion	Upload insurance claims CSV; validate columns; handle missing values; drop null-heavy columns (_c39); impute missing categorical fields (authorities_contacted)
FR-2	Data Preprocessing	Label encode categorical features; apply StandardScaler; split into 70/30 train-test sets; log-transform skewed features (policy_annual_premium)
FR-3	Exploratory Data Analysis	Generate countplot for fraud_reported; pie chart for incident_severity; histogram for age distribution; correlation heatmap; IQR-based outlier detection
FR-4	Model Training	Train Decision Tree Classifier; train Random Forest (100 estimators); train KNN (k=5); train Gradient Boosting Classifier
FR-5	Model Evaluation	Compute accuracy, F1-score, confusion matrix, classification report per model; compare models side-by-side
FR-6	Fraud Prediction	Accept new claim input; preprocess using saved scaler; return fraud probability and binary prediction (fraud / not fraud)
FR-7	Model Persistence	Save trained model using pickle; load model for inference without retraining
FR-8	Reporting & Dashboard	Display model comparison results; show feature importance plot; surface flagged claims for investigator review

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description

NFR-1	Usability	The system shall provide a clear, intuitive interface for investigators to upload claim data and view fraud predictions with minimal training required.
NFR-2	Security	All data transmissions shall use HTTPS/TLS. Access to the prediction system shall be role-based (investigator, admin). Sensitive PII in claims data shall be masked.
NFR-3	Reliability	The fraud detection model shall achieve a minimum F1-score of 0.80 on test data. The system shall handle malformed input gracefully with appropriate error messages.
NFR-4	Performance	Prediction for a single claim shall complete in under 500ms. Batch processing of 1000 claims shall complete within 60 seconds.
NFR-5	Availability	The system shall target 99.5% uptime. Deployments shall use rolling updates to ensure zero downtime.
NFR-6	Scalability	The architecture shall support horizontal scaling to handle increased claim volumes. The model retraining pipeline shall accommodate growing datasets without redesign.

Project Design Phase II

Technology Stack (Architecture & Stack)

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Table 1 – Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based dashboard for uploading claims CSV and viewing fraud predictions	Flask/HTML5, CSS3, Bootstrap, Jinja2
2	Data Ingestion Layer	Accepts CSV uploads, validates structure, triggers preprocessing pipeline	Python, Pandas
3	Data Preprocessing	Handles nulls, label encoding, scaling, outlier treatment	Scikit-learn (LabelEncoder, StandardScaler), NumPy
4	ML Model – Random Forest	Primary fraud classifier; 100 decision trees for ensemble prediction	Scikit-learn RandomForestClassifier
5	ML Model – Decision Tree	Interpretable rule-based classifier for explainability	Scikit-learn DecisionTreeClassifier
6	ML Model – KNN	Instance-based learner for anomaly proximity detection	Scikit-learn KNeighborsClassifier
7	ML Model – Gradient Boosting	Boosted ensemble for high-precision fraud detection	Scikit-learn GradientBoostingClassifier
8	Model Evaluation Module	Computes accuracy, F1-score, confusion matrix, classification report	Scikit-learn metrics, Matplotlib, Seaborn
9	Model Persistence	Serializes trained model for reuse without retraining	Python Pickle
10	Data Storage	Stores training dataset and intermediate processed data	Local Filesystem / CSV
11	Visualization Engine	Generates EDA charts: heatmap, histograms, countplots	Matplotlib, Seaborn
12	Infrastructure	Local Python environment; deployable to cloud (IBM Cloud / AWS)	Python 3.9+, Jupyter Notebook, Flask server

Table 2 – Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	All ML, data processing, and visualization libraries are fully open-source, ensuring no licensing costs	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Flask
2	Security Implementations	Role-based access control for the web dashboard; HTTPS for data transmission; input validation to prevent injection; PII masking on claim data display	Flask-Login, TLS/HTTPS, OWASP input validation
3	Scalable Architecture	Modular pipeline separating data ingestion, preprocessing, model training, and serving; each module independently scalable	Flask REST API, Modular Python packages
4	Availability	Stateless prediction API enables load-balanced horizontal scaling; model artifacts stored externally for fast startup	Gunicorn, Load Balancer (Nginx), Pickle model store
5	Performance	Preprocessing and prediction cached for repeated inputs; batch inference supported for bulk claim processing; lightweight models ensure sub-500ms latency	Pickle model caching, Pandas batch processing

Project Design Phase II

Data Flow Diagram & User Stories

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Data Flow Diagram – Level 0 (Context Diagram)

The system has three external entities: Insurance Investigator, Claims Database, and Admin. The core process is the AI Fraud Detection Engine.

External Entity	Data Flow Direction	Data Description
Insurance Investigator	→ AI Fraud Detection Engine	Uploads claim CSV file; requests prediction
AI Fraud Detection Engine	→ Insurance Investigator	Returns fraud probability score and prediction label
Claims Database (CSV)	→ AI Fraud Detection Engine	Historical labelled claim records (insurance_claims.csv)
AI Fraud Detection Engine	→ Data Store (Pickle)	Serialized trained ML model saved for inference
Data Store (Pickle)	→ AI Fraud Detection Engine	Loads trained model for real-time prediction
Admin	→ AI Fraud Detection Engine	Triggers model retraining; reviews system performance metrics
AI Fraud Detection Engine	→ Admin	Model accuracy, F1-score reports, confusion matrix

Data Flow Diagram – Level 1 (Process Decomposition)

Step	Process	Details
1	Data Ingestion	Investigator uploads CSV → System reads and validates columns → Drops _c39 (all-null column)
2	Preprocessing	Imputes missing values (mode for authorities_contacted) → Label encodes all categorical columns → Applies StandardScaler to numeric features
3	Feature Engineering	Drops target column (fraud_reported) to create feature matrix X → Retains target as y → Splits into x_train/x_test (70/30)

4	Model Training	Trains DecisionTree, RandomForest (n=100), KNN (k=5), GradientBoosting → Evaluates each on test set → Selects best model
5	Prediction Serving	New claim data preprocessed → Loaded pickle model predicts → Returns fraud/not-fraud label with probability
6	Reporting	Confusion matrix and classification report generated → Feature importance plotted → Results displayed on dashboard

User Stories

User Type	Epic	USN	User Story	Acceptance Criteria	Priority	Release
Investigator	Data Upload	USN-1	As an investigator, I can upload a CSV of insurance claims to the system	CSV is accepted; invalid files show error message	High	Sprint-1
Investigator	Data Upload	USN-2	As an investigator, I can see data validation results after upload (rows, columns, null count)	Summary stats displayed post-upload	High	Sprint-1
Investigator	Fraud Prediction	USN-3	As an investigator, I can click 'Predict' to get fraud probability for each claim	Each claim row shows fraud probability and label	High	Sprint-1
Investigator	Fraud Prediction	USN-4	As an investigator, I can filter results to show only flagged (high-risk) claims	Filtered view shows claims above threshold	Medium	Sprint-2
Investigator	Explainability	USN-5	As an investigator, I can view which features contributed most to a fraud prediction	Feature importance chart displayed per prediction	Medium	Sprint-2
Admin	Model Management	USN-6	As an admin, I can trigger model retraining with updated claims data	Model retrained and new accuracy metrics shown	High	Sprint-2
Admin	Model Management	USN-7	As an admin, I can view model performance metrics (accuracy, F1, confusion matrix)	Metrics dashboard accessible to admin role only	High	Sprint-1
Admin	User Management	USN-8	As an admin, I can add or remove investigator accounts	Role-based access enforced; new users can log in	Medium	Sprint-3

Investigator	Dashboard	USN-9	As an investigator, I can view a summary dashboard of daily fraud flags	Dashboard shows count of flagged vs cleared claims	Low	Sprint-3
Investigator	Export	USN-10	As an investigator, I can export the fraud prediction results to CSV	Downloaded CSV contains claim ID, prediction, probability	Low	Sprint-3